



## UOP10 – Laboratory steam generator

As many laboratories do not incorporate a centralised or reliable steam source, Armfield offers an advanced boiler providing a constant supply of steam. The compact unit requires only connection to water and electrical supplies for operation.

**PRODUCT CODE: UOP10**

**Categories:** Armfield Educational Food Technology, Educational, FT Series - Educational Food Technology, IFT Industrial Food Technology, UOP Series - Unit Operations, UOP10 - Laboratory Steam Generator

**Tags:** Advanced Boiler , Blow Down Separator , Continuous Steam Generator , Heat And Mass Transfer , Laboratory Steam Generator , Quick Start Up , Service Unit , Simple Controls , Steam , Unit Operations , UOP Series , UOP10 , Variable Pressure Outputs

### DESCRIPTION

As many laboratories do not incorporate a centralised or reliable steam source, Armfield offers an advanced boiler providing a constant supply of steam. The compact unit requires only connection to water and electrical supplies for operation.

The Armfield UOP10 Laboratory Steam Generator comprises a mains fed header tank with level control, which supplies the pressure vessel via a quiet running, heavy duty, high pressure feed pump.

The pressure vessel is built to all relevant British and European standards. Being heavily insulated and featuring two totally immersed Incoloy heating elements, it provides a highly efficient and economical means of generating a continuous supply of steam.

The two heating elements are individually switched and the unit will operate satisfactorily on a single element.

The integral control panel houses an on/off mains switch with switchable test functions, LEDs display the operational status of the equipment.

The UOP10 is fully safeguarded via a Magnetic Circuit Breaker (MCB) and a Residual Current Device (RCD) to give a long trouble free life.

The components of the UOP10 are mounted within a steel cabinet finished with a high gloss stove enamel paint for corrosion resistance and long life. Front panels are hinged for easy access and are fitted with key operated handles.

It is recommended that for optimum life of the unit, feedwater should be demineralised.

## TECHNICAL SPECIFICATIONS

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**Steam output at 100°C** – 55Kg/hr (continuous)

**Heat output** – 36kW

**Number of heating elements** – 2

**Max. operating pressure** – 10 bar

## FEATURES & BENEFITS

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- Simple controls allow for quick start up and shut down procedures
- Variable pressure outputs
- Rapid start up to full pressure and will deliver steam during heating process
- Automatic top up of feed tank and boiler during normal use
- Normal domestic water supply sufficient for normal operation
- Low water and high pressure alarms cut power to heating element to prevent unsafe conditions
- Dual locks to prevent unauthorised access to boiler internal controls
- Lock out isolation switch on front of unit for safety during maintenance
- Rapid and variable blowdown of steam boiler during shutdown of system
- Pressure relief valves in case of over-pressurisation to prevent energetic discharge

## Ordering Specification

- A floor standing continuous steam generator
- Splash proof electrical enclosure and switches
- Solid state level control on feed water tank
- Easy and safe access for maintenance and inspection, with door isolator to the main electrical enclosure
- Manufactured in accordance with Module G of the European
- Pressure Equipment directive 97/23/EC
- A comprehensive instruction manual

## Optional Accessories

In some countries it is illegal to blow down a steam boiler directly into the drainage system and a separator must be used. Armfield can offer a purpose built separator to fulfil this function. Users should check local regulations on the requirements for this unit.

**UOP10-10:** Blow Down Separator

## Requirements

- **Electrical supply:** Three phase (see ordering codes)
- **Water supply:** 1.0l/min @ 2 bar maximum usage

## Shipping Specifications



# PACKED AND CRATED SHIPPING SPECIFICATIONS

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**Volume:** 2.1m<sup>3</sup>

**Gross Weight:** 490Kg

## Dimensions

**Length:** 1.12m

**Width:** 0.68m

**Height:** 1.92m

**Note:** Total height of 2.5m is required for access

## Order Codes

- **UOP10-C:** 415V/3ph/50Hz, 57A
- **UOP10-D:** 208V/3ph/60Hz, 112A
- **UOP10-E:** 380V/3ph/50Hz, 57A
- **UOP10-F:** 220V/3ph/60Hz, 112A
- **UOP10-10:** Blow Down Separator

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